

SKILLWEEK4

TASK1: To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output and Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures

```
jhansi_priya@Jhansi-Priya: ~/ skill4
jhansi_priya@Jhansi-Priya:~/skill4$ nano task1.c
jhansi_priya@Jhansi-Priya:~/skill4$ gcc task1.c -o task1
jhansi_priya@Jhansi-Priya:~/skill4$ ./task1
Enter a command: echo Hello World
Argument 1: echo
Argument 2: Hello
Argument 3: World
jhansi_priya@Jhansi-Priya:~/skill4$ |
```

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    char *token = strtok(input, " \\t\\n");

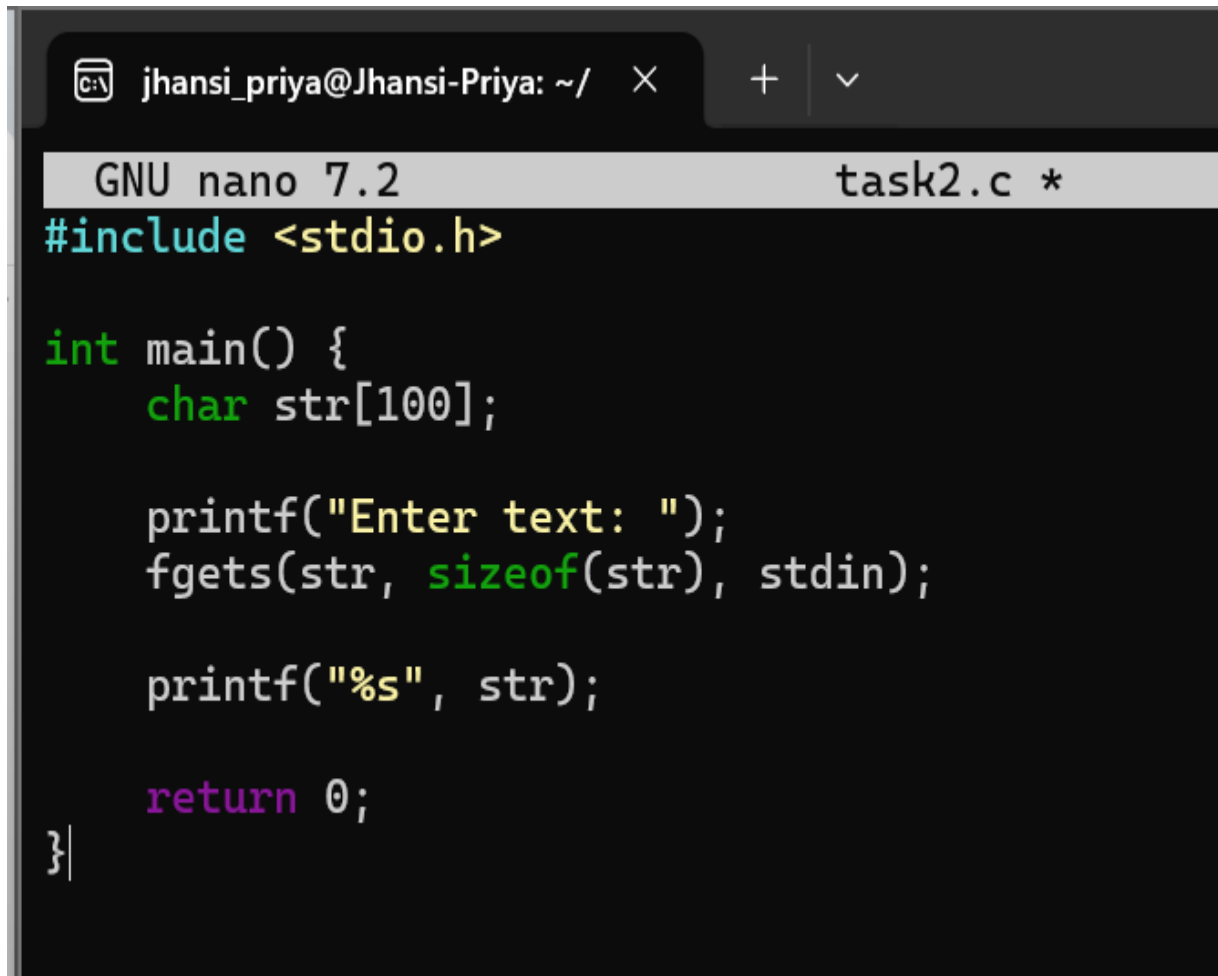
    int count = 1;

    while (token != NULL) {
        printf("Argument %d: %s\\n", count++, token);
        token = strtok(NULL, " \\t\\n");
    }

    return 0;
}
```

TASK2: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

```
Argument 3: World
jhansi_priya@Jhansi-Priya:~/skill4$ nano task2.c
jhansi_priya@Jhansi-Priya:~/skill4$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/skill4$ ./task2
Enter text: echo "Hello World"
echo "Hello World"
```



```
GNU nano 7.2 task2.c *
#include <stdio.h>

int main() {
    char str[100];

    printf("Enter text: ");
    fgets(str, sizeof(str), stdin);

    printf("%s", str);

    return 0;
}
```

TASK3: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

```
echo "Hello World"
jhansi_priya@Jhansi-Priya:~/skill4$ echo "Hello $USER"
Hello jhansi_priya
jhansi_priya@Jhansi-Priya:~/skill4$ |
```

TASK4: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

```

Hetto jhansi_priya
jhansi_priya@Jhansi-Priya:~/skill4$ mkdir My\skill4
jhansi_priya@Jhansi-Priya:~/skill4$ touch task4.txt
jhansi_priya@Jhansi-Priya:~/skill4$ echo \$HOME
$HOME
jhansi_priya@Jhansi-Priya:~/skill4$ |
```

TASK5: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```

jhansi_priya@Jhansi-Priya:~/skill4$ nano task5.c
jhansi_priya@Jhansi-Priya:~/skill4$ gcc task5.c -o task5
jhansi_priya@Jhansi-Priya:~/skill4$ ./task5
Parent PID: 3559
Child PID: 3560
Myskill4 task1.c task2.c task5
task1 task2 task4.txt task5.c
Child process completed.
jhansi_priya@Jhansi-Priya:~/skill4$ |
```

```

GNU nano 7.2 task5.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        execlp("ls", "ls", NULL);
    } else {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }

    return 0;
}
```

